

# GradeWise Service

Automated Exam Evaluation using AI

✓ Question Paper > ✓ Model Answer > ✓ Student Sheet > 4 Results

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## Evaluation Result

Student ID: MH004

P

27.0 / 40.0

### Scoreboard

| Question  | Score | Max | %    |     | Status |
|-----------|-------|-----|------|-----|--------|
| Section A |       |     |      |     |        |
| Q1        | 1     | 1   | 100% | 1/1 | Scored |
| Q2        | 1     | 1   | 100% | 1/1 | Scored |
| Q3        | 1     | 1   | 100% | 1/1 | Scored |
| Q4        | 1     | 1   | 100% | 1/1 | Scored |
| Q5        | 1     | 1   | 100% | 1/1 | Scored |
| Q6        | 1     | 1   | 100% | 1/1 | Scored |
| Q7        | 1     | 1   | 100% | 1/1 | Scored |

| Question  | Score | Max | %    |       | Status     |
|-----------|-------|-----|------|-------|------------|
| Q8        | 1     | 1   | 100% | 1/1   | Scored     |
| Section B |       |     |      |       |            |
| Q9        | 2     | 2   | 100% | 2/2   | Scored     |
| Q10       | 2     | 2   | 100% | 2/2   | Scored     |
| Q11       | 2     | 2   | 100% | 2/2   | Scored     |
| Q12       | 2     | 2   | 100% | 2/2   | Scored     |
| Q13       | 2     | 2   | 100% | 2/2   | Scored     |
| Q14i      | 0     | 1   | 0%   | 0/1   | Not Scored |
| Q14ii     | 0     | 1   | 0%   | 0/1   | Not Scored |
| Section C |       |     |      |       |            |
| Q15       | 0.5   | 3   | 17%  | 0.5/3 | Scored     |
| Q16i      | 0     | 1.5 | 0%   | 0/1.5 | Not Scored |
| Q16ii     | 0     | 1.5 | 0%   | 0/1.5 | Not Scored |
| Q17       | 3     | 3   | 100% | 3/3   | Scored     |
| Q18       | 1.5   | 3   | 50%  | 1.5/3 | Scored     |
| Section D |       |     |      |       |            |
|           |       |     |      |       |            |

| Question | Score | Max | %   | Status                  |
|----------|-------|-----|-----|-------------------------|
| Q19      | 4     | 8   | 50% | 4/8 <span>Scored</span> |
| Total    | 27    | 40  | 68% | —                       |

## Question-wise Details

### Question 1

1 / 1

The student's answer matches the model answer exactly. The number of subsets of a set with 3 elements is  $2^3 = 8$ , which corresponds to option (b). Therefore, full marks are awarded.

### Question 2

1 / 1

The student's answer matches the model answer exactly, correctly identifying the universal set as the set which contains all elements under consideration. Therefore, full marks are awarded.

### Question 3

1 / 1

The student's answer matches the model answer exactly. Since  $\sin A = 3/5$  and  $A$  is acute,  $\cos A = 4/5$  is correct, corresponding to option (a). Full marks are awarded.

#### 🚩 Suggestions

- No improvements needed as the answer is correct and complete.

### Question 4

1 / 1

The student's answer '(c) Many-one relation' matches the model answer '(c) Many-one relation' exactly in meaning and choice, with only a minor typographical difference in the dash character which does not affect correctness. Therefore, full marks are awarded.

 **Suggestions**

- Ensure consistent use of dash characters (en dash vs hyphen) for better presentation.

**Question 5**

1 / 1

The student's answer '(b)  $\frac{1}{2}$ ' is equivalent to the model answer '(b)  $\frac{1}{2}$ '. Both represent the same value and option choice, so full marks are awarded.

 **Suggestions**

- The student should continue to provide answers clearly and can use either ' $\frac{1}{2}$ ' or ' $\frac{1}{2}$ ' as both are acceptable and equivalent.

**Question 6**

1 / 1

The student's answer matches the model answer exactly, selecting option (c) 60, which is the correct number of ways to arrange 3 books out of 5 different books.

**Question 7**

1 / 1

The student's answer exactly matches the model answer, correctly identifying that if A and B are disjoint sets, then their intersection is the empty set ( $A \cap B = \emptyset$ ).

 **Suggestions**

- No improvements needed as the answer is correct and complete.

**Question 8**

1 / 1

The student's answer matches the model answer exactly, correctly identifying the relation 'less than' on the set of natural numbers as transitive.

 **Suggestions**

- No improvements needed as the answer is correct and complete.

### Question 9

2 / 2

The student's answer correctly defines a set as a well-defined collection of objects and provides a valid example  $A = \{1, 2, 3\}$ , matching the model answer completely.

#### 🚩 Suggestions

- None. The answer is complete and accurate.

### Question 10

2 / 2

The student correctly identified the domain as all real numbers and the range as  $[0, \infty)$ , which matches the model answer's domain ( $\mathbb{R}$ ) and range ( $x \geq 0$ ). The notation used by the student is clear and mathematically equivalent to the model answer.

#### 🚩 Suggestions

- Use standard mathematical notation for domain, such as  $\mathbb{R}$ , to improve clarity.
- Explicitly state the domain and range using set notation if possible, e.g.,  
Domain =  $\{x \in \mathbb{R}\}$ , Range =  $\{y \in \mathbb{R} : y \geq 0\}$ .

### Question 11

2 / 2

The student correctly evaluated  $\sin 45^\circ \cos 45^\circ$  by substituting the values  $(1/\sqrt{2})$  for both sine and cosine, and correctly simplified the product to  $1/2$ . The answer matches the model answer both in method and final value.

#### 🚩 Suggestions

- Ensure to include spaces between function names and angles for clarity, e.g., ' $\sin 45^\circ$ ' instead of ' $\sin 45^\circ$ '.
- Consider writing intermediate steps explicitly to enhance readability, although the current steps are sufficient.

### Question 12

2 / 2

The student correctly calculated the value as 20 with the correct multiplication steps shown. Although the notation ' ${}^5P_2$ ' is incorrect for the question ' ${}^5P_4$ ', the numerical value and calculation steps correspond to the correct value of  ${}^5P_4$ , which is  $5 \times 4 \times 3 \times 2 = 120$ . However, the student's calculation only multiplies  $5 \times 4 = 20$ , which is the value of  ${}^5P_2$ , not  ${}^5P_4$ . Therefore, the final numerical value is incorrect for the question asked.

🚩 **Suggestions**

- Ensure that you understand the permutation notation and calculate the correct number of terms for  ${}^5P_4$ , which involves multiplying  $5 \times 4 \times 3 \times 2$ .
- Double-check the final numerical value to match the question requirements.
- Use correct permutation notation to avoid confusion.

### Question 13

**2 / 2**

The student correctly identified that the number of elements in the power set of a set with 4 elements is  $2^4 = 16$ . The final numerical value matches the model answer, and the student showed the correct formula and calculation.

🚩 **Suggestions**

- The student could improve clarity by explicitly stating that the power set contains all subsets of the original set.
- Including a brief explanation of why the power set has  $2^n$  elements (where  $n$  is the number of elements in the original set) would demonstrate deeper understanding.

### Question 14i

**0 / 1**

No student answer available.

🚩 **Suggestions**

- Provide an answer to receive credit.

### Question 14ii

**0 / 1**

No student answer available.

 **Suggestions**

- Provide an answer to receive credit.

**Question 15**

0.5 / 3

The student correctly identifies the left-hand side (LHS) and right-hand side (RHS) of the identity and mentions taking the LCM and simplifying to reach the RHS. However, the student does not show any intermediate algebraic steps or detailed working to prove the identity. The final conclusion is stated without justification or demonstration of the key algebraic manipulations. Therefore, the answer is incomplete and lacks the necessary proof steps, but the final conclusion is correct.

 **Suggestions**

- Show all intermediate steps clearly, including taking the LCM, expanding terms, and simplifying the numerator and denominator.
- Demonstrate how  $\sin^2\theta + (1 + \cos\theta)^2$  simplifies to  $2 + 2\cos\theta$ .
- Explain how the expression reduces to  $2 / \sin\theta$  and then to  $2 \operatorname{cosec}\theta$ .
- Avoid skipping steps to ensure the proof is complete and easy to follow.

**Question 16i**

0 / 1.5

No student answer available.

 **Suggestions**

- Provide an answer to receive credit.

**Question 16ii**

0 / 1.5

No student answer available.

 **Suggestions**

- Provide an answer to receive credit.

### Question 17

3 / 3

The student's answer correctly calculates the number of 3-digit numbers without repetition using the digits 1, 2, 3, 4, 5. The final numerical value, 60, matches the model answer, and the steps are clearly shown. Although the notation '5P3' is used instead of the model's ' ${}^5P_3$ ', this is a standard and acceptable permutation notation.

#### 🚩 Suggestions

- Maintain clarity by explicitly stating the permutation formula used, such as  $nPr = n \times (n-1) \times \dots$
- Ensure consistent notation to avoid confusion, but minor notation differences are acceptable.

### Question 18

1.5 / 3

The student correctly identified that the relation is reflexive, symmetric, and transitive, which matches the model answer's conclusion. However, the student did not provide any explanation or justification for these properties. Since the final conclusion is correct but lacks supporting reasoning, half marks are awarded.

#### 🚩 Suggestions

- Provide explanations or justifications for why the relation is reflexive, symmetric, and transitive.
- Show that  $(a,a) \in R$  for all elements  $a$  in  $A$  to prove reflexivity.
- Explain why for every  $(x,y)$  in  $R$ ,  $(y,x)$  is also in  $R$  to prove symmetry.
- Demonstrate that for any  $(x,y)$  and  $(y,z)$  in  $R$ ,  $(x,z)$  is also in  $R$  to prove transitivity.

### Question 19

4 / 8

The student correctly stated the key identity  $1 + \tan^2\theta = \sec^2\theta$  and concluded that the left-hand side equals the right-hand side, thus proving the identity. However, the student did not explicitly show the substitution step or the intermediate algebraic manipulation as in the model answer. The reasoning is correct but lacks detailed working steps.

#### 🚩 Suggestions



- Include the substitution step explicitly: replace  $(1 + \tan^2\theta)$  with  $\sec^2\theta$  in the expression.
- Show the intermediate step where LHS becomes  $\sec^2\theta(1 + \sin^2\theta)$  before concluding equality with RHS.
- Provide a brief explanation of why the identity  $1 + \tan^2\theta = \sec^2\theta$  holds, if possible, to strengthen the proof.